

15.C57 Optimization Methods HW1

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Question 1

(a)

True.

Since A has m linearly independent rows, $\text{rank}(A) = m$. Since there are m independent equality constraints, they reduce the n variables to $n - m$ degrees of freedom.

$$n - \text{rank}(A) = n - m = (m + 1) - m = 1.$$

Thus, $Ax = b$ describes a line. Intersecting this line with our nonnegativity constraints $x \geq 0$ gives us a feasible region with at most two extreme points. Since basic feasible solutions correspond to extreme points, $\mathcal{P}$ has at most two basic feasible solutions. $\square$

(b)

False. A basic feasible solution can have at most m positive variables, but an optimal solution does not have to be a basic feasible solution; it may lie between extreme points when there are multiple optimal solutions.

Consider an example with $m = 2$ constraints and $n = 3$ variables:

$$x_1 + x_2 + x_3 = 3,$$

$$x_1 - x_2 = 0,$$

with

$$x_1, x_2, x_3 \geq 0.$$

Let

$$c = (0, 0, 0)^\top.$$

This makes our objective

$$\min c^\top x = \min(0x_1 + 0x_2 + 0x_3) = 0.$$

Therefore, every feasible solution is optimal.

For example,

$$x = (1, 1, 1)^\top$$

is feasible and optimal. However, all three variables are positive, while $m = 2$. Thus, this optimal solution has more than m positive variables, so the statement is false.

(c)

True.

If the linear optimization problem has an optimal solution, then there exists an optimal solution x^* that is an extreme point of $\mathcal{P}$. Since extreme points are equivalent to basic feasible solutions, x^* is also a basic feasible solution.

For a standard-form polyhedron, a basic feasible solution has $n - m$ nonbasic variables set to zero. Therefore, at most m variables can be positive. Hence, there exists an optimal solution with at most m positive variables.

(d)

True.

By the theorem from L2 slide 11, a point is a basic feasible solution if and only if it is an extreme point of $\mathcal{P}$. Thus, x and y are two distinct extreme points of $\mathcal{P}$.

For any $\lambda \in (0, 1)$, let

$$z = \lambda x + (1 - \lambda)y.$$

Since $0 < \lambda < 1$ and $x \neq y$, the point z lies between the two distinct feasible points x and y . By the definition of an extreme point, z cannot be an extreme point of $\mathcal{P}$.

Therefore, by the equivalence between extreme points and basic feasible solutions, z is not a basic feasible solution.

(e)

True.

Let x_j be a nonbasic variable with zero reduced cost,

$$\bar{c}_j = 0.$$

Since x is nondegenerate, all basic variables are strictly positive. Therefore, for the direction d associated with increasing x_j , we can choose a sufficiently small $\theta > 0$ such that

$$y = x + \theta d$$

is still feasible. Since $\theta > 0$, we have $y \neq x$.

The objective value at y is

$$c^\top y = c^\top (x + \theta d) = c^\top x + \theta \bar{c}_j = c^\top x.$$

Thus, y has the same objective value as x . Since x is optimal, y is also optimal. Therefore, there are at least two distinct optimal solutions, so x is not the unique optimal solution.

(f)

False.

The function $\max\{c^\top x, d^\top x\}$ is convex, since it is the maximum of two linear functions, but it is not necessarily linear. Therefore, the theorem that a linear optimization problem has an optimal solution at an extreme point does not necessarily apply.

Consider the one-dimensional polyhedron

$$\mathcal{P} = [-1, 1].$$

Its extreme points are -1 and 1 .

Let

$$c = 1 \quad \text{and} \quad d = -1.$$

$$c^\top x = x \quad \text{and} \quad d^\top x = -x.$$

Therefore, our objective becomes

$$\max\{c^\top x, d^\top x\} = \max\{x, -x\} = |x|.$$

Hence, our problem to minimize is

$$\min_{x \in [-1, 1]} |x|.$$

The minimum is attained at

$$x^* = 0,$$

with objective value 0.

However, 0 is not an extreme point of $\mathcal{P}$, since it lies between the two extreme points -1 and 1 . At either extreme point, the objective value is 1.

Thus, the problem has an optimal solution, but no optimal solution is an extreme point of $\mathcal{P}$. Therefore, the statement is false.

Question 2

The feasible region is determined by

$$x_1 + x_2 \leq 5,$$

$$2x_1 + x_2 \leq 8,$$

$$x_1 + 3x_2 \leq 9,$$

together with $x_1, x_2 \geq 0$.

(a)

We first find the extreme points on the coordinate axes. Setting $x_2 = 0$, the constraints imply

$$x_1 \leq 5, \quad x_1 \leq 4, \quad x_1 \leq 9,$$

so the extreme point on the x_1 -axis is

$$(4, 0).$$

Similarly, setting $x_1 = 0$, the constraints imply

$$x_2 \leq 5, \quad x_2 \leq 8, \quad x_2 \leq 3,$$

so the extreme point on the x_2 -axis is

$$(0, 3).$$

To find the remaining extreme point, consider the intersection of the first two constraints:

$$x_1 + x_2 = 5,$$

$$2x_1 + x_2 = 8.$$

Subtracting the first equation from the second gives us

$$x_1 = 3, \quad x_2 = 2. \quad (3, 2).$$

$$3 + 3(2) = 9,$$

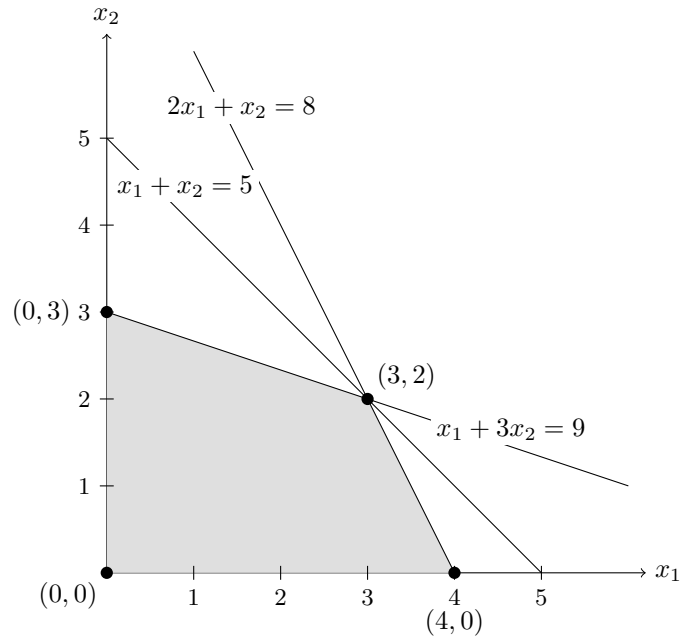
So the third constraint is also active at this point.

We summarize:

Extreme point	Active inequality constraints
(0, 0)	None
(4, 0)	$2x_1 + x_2 = 8$
(3, 2)	$x_1 + x_2 = 5, 2x_1 + x_2 = 8, x_1 + 3x_2 = 9$
(0, 3)	$x_1 + 3x_2 = 9$

The feasible region has extreme points

$$(0, 0), \quad (4, 0), \quad (3, 2), \quad (0, 3).$$



(b)

We rewrite with slack variables $s_1, s_2, s_3 \geq 0$, so the problem in standard form is

$$\max \quad x_1 + \lambda x_2$$

subject to

$$x_1 + x_2 + s_1 = 5,$$

$$2x_1 + x_2 + s_2 = 8,$$

$$x_1 + 3x_2 + s_3 = 9,$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0.$$

There are three equality constraints, so each basis contains three variables. For each extreme point, the corresponding slack variables and one possible basis are:

(x_1, x_2)	$(x_1, x_2, s_1, s_2, s_3)$	One basis
(0, 0)	(0, 0, 5, 8, 9)	$\{s_1, s_2, s_3\}$
(4, 0)	(4, 0, 1, 0, 5)	$\{x_1, s_1, s_3\}$
(0, 3)	(0, 3, 2, 5, 0)	$\{x_2, s_1, s_2\}$
(3, 2)	(3, 2, 0, 0, 0)	$\{x_1, x_2, s_1\}$

The BFS corresponding to (3, 2) is degenerate. Since there are three equality constraints, a basis must contain three variables. However, at (3, 2),

$$s_1 = s_2 = s_3 = 0.$$

So any basis corresponding to this point must contain a zero-valued basic variable. For example, with basis

$$\{x_1, x_2, s_1\},$$

the basic variables have values

$$(x_1, x_2, s_1) = (3, 2, 0).$$

Therefore, this BFS is degenerate.

The other three BFSs are nondegenerate since all basic variables in the displayed bases are strictly positive.

Geometrically, $(3, 2)$ is degenerate because all three inequality constraints are active at this point:

$$x_1 + x_2 = 5, \quad 2x_1 + x_2 = 8, \quad x_1 + 3x_2 = 9.$$

Since the problem is two-dimensional, only two linearly independent active constraints are needed to determine an extreme point. As we have three active constraints at $(3, 2)$, this corresponds to degeneracy.

(c)

The objective function is

$$x_1 + \lambda x_2.$$

At each extreme point, we see:

Extreme point	Objective value
$(0, 0)$	0
$(4, 0)$	4
$(3, 2)$	$3 + 2\lambda$
$(0, 3)$	3λ

The origin will never "win" because $(4, 0)$ always gives $4 > 0$. We first compare $(4, 0)$ and $(3, 2)$:

$$4 = 3 + 2\lambda, \quad \lambda = \frac{1}{2}.$$

Next, compare $(3, 2)$ and $(0, 3)$:

$$3 + 2\lambda = 3\lambda, \quad \lambda = 3.$$

We can see that for values of λ away from these breakpoints,

$$x^*(\lambda) = \begin{cases} (4, 0), & \lambda < \frac{1}{2}, \\ (3, 2), & \frac{1}{2} < \lambda < 3, \\ (0, 3), & \lambda > 3. \end{cases}$$

We also observe that multiple optima occur at $\lambda = \frac{1}{2}$ and $\lambda = 3$.

At $\lambda = \frac{1}{2}$, the objective becomes

$$x_1 + \frac{1}{2}x_2.$$

Dividing constraint (2) by 2 gives

$$x_1 + \frac{1}{2}x_2 \leq 4.$$

So the objective value is at most 4, and every feasible point satisfying

$$2x_1 + x_2 = 8$$

attains this value.

Therefore, the entire feasible edge from $(4, 0)$ to $(3, 2)$ is optimal. Hence the complete set of optimal solutions is the line segment joining $(4, 0)$ and $(3, 2)$:

$$\{t(4, 0) + (1 - t)(3, 2) : 0 \leq t \leq 1\}.$$

At $\lambda = 3$, the objective is

$$x_1 + 3x_2.$$

Since

$$x_1 + 3x_2 \leq 9,$$

every point on the feasible edge

$$x_1 + 3x_2 = 9$$

has objective value 9. So we get our complete set of optimal solutions, which is the line segment joining $(3, 2)$ and $(0, 3)$:

$$\{t(3, 2) + (1 - t)(0, 3) : 0 \leq t \leq 1\}.$$

Thus,

$$\operatorname{argmax} = \begin{cases} \{(4, 0)\}, & \lambda < \frac{1}{2}, \\ [(4, 0), (3, 2)], & \lambda = \frac{1}{2}, \\ \{(3, 2)\}, & \frac{1}{2} < \lambda < 3, \\ [(3, 2), (0, 3)], & \lambda = 3, \\ \{(0, 3)\}, & \lambda > 3. \end{cases}$$

The problem has multiple optimal solutions precisely when

$$\lambda = \frac{1}{2} \quad \text{or} \quad \lambda = 3.$$

(d)

For $\lambda = 1$, the problem in standard form is

$$\max \quad x_1 + x_2$$

subject to

$$x_1 + x_2 + s_1 = 5,$$

$$2x_1 + x_2 + s_2 = 8,$$

$$x_1 + 3x_2 + s_3 = 9,$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0.$$

We use:

$$(x_1, x_2, s_1, s_2, s_3),$$

we have

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 1 & 3 & 0 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ 8 \\ 9 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

For this maximization problem, we use the reduced costs

$$\bar{c}^\top = c^\top - c_B^\top B^{-1}A.$$

A positive reduced cost indicates an improving direction.

Initial BFS.

Starting with $x_1 = x_2 = 0$ gives us

$$(x_1, x_2, s_1, s_2, s_3) = (0, 0, 5, 8, 9).$$

The initial basis is

$$\{s_1, s_2, s_3\},$$

so

$$B = I, \quad B^{-1} = I, \quad c_B = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore,

$$\bar{c}^\top = c^\top - c_B^\top B^{-1} A = (1 \quad 1 \quad 0 \quad 0 \quad 0).$$

Both x_1 and x_2 have positive reduced cost. We choose x_1 to enter the basis.

The entering column is

$$A_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

The direction of the basic variables is

$$d_B = -B^{-1} A_1 = \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix}.$$

Thus,

$$s_1 = 5 - \theta, \quad s_2 = 8 - 2\theta, \quad s_3 = 9 - \theta.$$

The minimum-ratio test gives us

$$\theta^* = \min \left\{ \frac{5}{1}, \frac{8}{2}, \frac{9}{1} \right\} = 4.$$

Therefore, we let s_2 leaves the basis and the new BFS is

$$(x_1, x_2, s_1, s_2, s_3) = (4, 0, 1, 0, 5).$$

Second iteration.

The new basis is

$$\{s_1, x_1, s_3\}.$$

Thus,

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

$$B^{-1} = \begin{pmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 \end{pmatrix}.$$

Since the basic variables are (s_1, x_1, s_3) ,

$$c_B = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

$$c_B^\top B^{-1} = (0 \quad \frac{1}{2} \quad 0),$$

$$c_B^\top B^{-1}A = \begin{pmatrix} 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}.$$

$$\bar{c}^\top = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 \end{pmatrix}.$$

Since

$$\bar{c}_{x_2} = \frac{1}{2} > 0,$$

x_2 enters the basis.

The entering column is

$$A_2 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}.$$

The direction of the basic variables is

$$d_B = -B^{-1}A_2 = -\begin{pmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ -\frac{5}{2} \end{pmatrix}.$$

Since the basic variables are ordered as (s_1, x_1, s_3) , we have

$$s_1 = 1 - \frac{1}{2}\theta,$$

$$x_1 = 4 - \frac{1}{2}\theta,$$

$$s_3 = 5 - \frac{5}{2}\theta.$$

The minimum-ratio test gives

$$\theta^* = \min \left\{ \frac{1}{1/2}, \frac{4}{1/2}, \frac{5}{5/2} \right\} = \min\{2, 8, 2\} = 2.$$

There is a tie between s_1 and s_3 . Choose s_1 to leave the basis. At $\theta = 2$, the new BFS is

$$(x_1, x_2, s_1, s_2, s_3) = (3, 2, 0, 0, 0).$$

With s_1 leaving and x_2 entering, choose the basis

$$\{x_2, x_1, s_3\}.$$

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 3 & 1 & 1 \end{pmatrix},$$

$$B^{-1} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 1 & 0 \\ -5 & 2 & 1 \end{pmatrix}.$$

$$c_B = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

$$c_B^\top B^{-1} = (1 \quad 0 \quad 0),$$

$$c_B^\top B^{-1} A = (1 \quad 1 \quad 1 \quad 0 \quad 0).$$

$$\bar{c}^\top = (1 \quad 1 \quad 0 \quad 0 \quad 0) - (1 \quad 1 \quad 1 \quad 0 \quad 0) = (0 \quad 0 \quad -1 \quad 0 \quad 0).$$

The nonbasic variables are s_1 and s_2 , with

$$\bar{c}_{s_1} = -1, \quad \bar{c}_{s_2} = 0.$$

Since all nonbasic reduced costs are nonpositive, there is no improving direction. Therefore the current BFS is optimal.

$$\boxed{x_1^* = 3, \quad x_2^* = 2,}$$

with optimal objective value

$$z^* = 3 + 2 = 5.$$

The simplex iterations we get:

$$\boxed{(0, 0, 5, 8, 9) \longrightarrow (4, 0, 1, 0, 5) \longrightarrow (3, 2, 0, 0, 0).}$$

First iteration: x_1 enters, s_2 leaves, and $\theta^* = 4$.

Second iteration: x_2 enters, there is a tie between s_1 and s_3 for the leaving variable, and $\theta^* = 2$. We choose s_1 to leave which yields the optimal BFS above.

(e)

In the second ratio test, both s_1 and s_3 become zero at

$$\theta = 2.$$

This means that two constraints become tight at the same time when we reach

$$(x_1, x_2) = (3, 2).$$

Since only one variable leaves the basis, the other one stays basic with value zero. Therefore, the new BFS is **degenerate**.

However, this does not mean that simplex fails to make progress. In this step, we have

$$\theta = 2 > 0,$$

and the objective value increases from 4 to 5. Thus, we still move to a better solution.

Degeneracy can cause a later pivot to have $\theta = 0$. In our case, after reaching $(3, 2)$, one of the basic variables, s_3 , is already equal to zero. If a later pivot requires s_3 to decrease, the ratio test would give $\theta = 0$ since s_3 cannot become negative. The basis could then change without changing the actual solution or objective value. However, this does not happen in the iteration above, since we reach $(3, 2)$ with a positive step size $\theta = 2$ and the objective increases from 4 to 5.

Question 3

(a)

Let

$$B = [A_{B(1)}, \dots, A_{B(m)}]$$

be the matrix formed by the m basic columns of A . By assumption, these columns are linearly independent. Since B is an $m \times m$ matrix, it follows that B is invertible.

There are $n - m$ nonbasic variables. We partition these variables into the sets L and U , and fix them at their bounds:

$$x_j = 0 \quad \text{for } j \in L,$$

and

$$x_j = u_j \quad \text{for } j \in U.$$

Therefore, each of the $n - m$ nonbasic variables has an active bound constraint.

We now consider the equality constraints

$$Ax = b.$$

When we separate the basic and nonbasic variables, we get

$$Bx_B + \sum_{j \in L} A_j x_j + \sum_{j \in U} A_j x_j = b.$$

Since $x_j = 0$ for $j \in L$ and $x_j = u_j$ for $j \in U$, this becomes

$$Bx_B + \sum_{j \in U} A_j u_j = b.$$

$$Bx_B = b - \sum_{j \in U} A_j u_j.$$

Since B is invertible, we can solve uniquely for the basic variables:

$$x_B = B^{-1} \left(b - \sum_{j \in U} A_j u_j \right).$$

Hence, the $n - m$ active bound constraints uniquely determine all of the nonbasic variables, while the m equality constraints uniquely determine the remaining m basic variables. Therefore, all n variables are uniquely determined by these active constraints.

In other words, these active constraints have full rank n . By the definition of a basic solution (active constraints span the whole space), the resulting vector x is therefore a basic solution. $\square$

For the nondegeneracy condition, recall that the $n - m$ nonbasic variables are fixed at either their lower or upper bounds. Thus, their $n - m$ bound constraints are active. Together with the m equality constraints $Ax = b$, these give the n active constraints defining the basic solution.

Suppose first that every basic variable satisfies

$$x_j \neq 0 \quad \text{and} \quad x_j \neq u_j.$$

Since $0 \leq x_j \leq u_j$, this is equivalent to

$$0 < x_j < u_j.$$

So no basic variable has an active bound constraint. Hence there are no extra active bound constraints beyond those defining the basic solution, so the basic solution is nondegenerate.

Conversely, suppose that x is nondegenerate. If some basic variable x_j satisfied

$$x_j = 0$$

or

$$x_j = u_j,$$

then its lower-bound or upper-bound constraint would also be active. This would give an additional active constraint beyond the m equality constraints and the $n - m$ active bound constraints of the nonbasic variables, making the basic solution degenerate.

This contradicts the assumption that x is nondegenerate. Therefore, no basic variable can lie at either of its bounds. Hence,

$$x_j \neq 0 \quad \text{and} \quad x_j \neq u_j$$

for every basic variable x_j .

Therefore,

$$x \text{ is nondegenerate} \iff x_j \neq 0 \text{ and } x_j \neq u_j \text{ for every basic variable } x_j. \quad \square$$

(b)

Since $x_j = 0$ and $\bar{c}_j < 0$, we increase x_j by θ :

$$x_j^{\text{new}} = \theta.$$

To maintain $Ax = b$, the basic variables are adjusted according to

$$x_B^{\text{new}} = x_B - \theta B^{-1} A_j.$$

Define

$$\boxed{d_B = -B^{-1} A_j.}$$

Then

$$x_B^{\text{new}} = x_B + \theta d_B.$$

For each basic variable x_i , let d_i denote its corresponding component of the direction vector d_B . Thus,

$$x_i^{\text{new}} = x_i + \theta d_i.$$

We now find the largest θ such that all variables remain within their bounds.

First, since the entering variable x_j starts at 0 and must satisfy $x_j \leq u_j$, we require

$$\theta \leq u_j.$$

For each basic variable x_i , we require

$$0 \leq x_i + \theta d_i \leq u_i.$$

If $d_i < 0$, then x_i decreases as θ increases. The lower bound $x_i + \theta d_i \geq 0$ therefore gives us

$$\theta \leq \frac{x_i}{-d_i}.$$

If $d_i > 0$, then x_i increases as θ increases. The upper bound $x_i + \theta d_i \leq u_i$ therefore gives us

$$\theta \leq \frac{u_i - x_i}{d_i}.$$

If $d_i = 0$, then x_i does not change and there's no restriction on θ .

So, our largest feasible step is

$$\theta^* = \min \left\{ u_j, \min_{i \in B: d_i < 0} \frac{x_i}{-d_i}, \min_{i \in B: d_i > 0} \frac{u_i - x_i}{d_i} \right\}.$$

The new basic columns depend on which term attains this minimum.

If $\theta^* = u_j$, then x_j reaches its upper bound before any basic variable reaches a bound. Thus, x_j moves from its lower bound to its upper bound u_j , and the basis does not change.

If the minimum is attained by some other basic variable x_i , then x_i reaches either 0 or u_i and leaves the basis. The entering variable x_j becomes basic, so the column A_j replaces A_i in the basis.

(c)

Since $x_j = u_j$ and $\bar{c}_j > 0$, we decrease x_j by θ :

$$x_j^{\text{new}} = u_j - \theta.$$

For $Ax = b$, we adjust the basic variables:

$$x_B^{\text{new}} = x_B + \theta B^{-1} A_j.$$

We define:

$$d_B = B^{-1} A_j.$$

So

$$x_B^{\text{new}} = x_B + \theta d_B.$$

Again, for each basic variable x_i , let d_i denote its corresponding component of the direction vector d_B . Thus,

$$x_i^{\text{new}} = x_i + \theta d_i.$$

Find the largest θ so that all variables remain within their bounds.

Since x_j starts at u_j and decreases according to $x_j^{\text{new}} = u_j - \theta$, we require

$$\theta \leq u_j.$$

For each basic variable x_i , we require

$$0 \leq x_i + \theta d_i \leq u_i.$$

If $d_i < 0$, then x_i decreases as θ increases. The lower bound $x_i + \theta d_i \geq 0$ therefore gives us

$$\theta \leq \frac{x_i}{-d_i}.$$

If $d_i > 0$, then x_i increases as θ increases. The upper bound $x_i + \theta d_i \leq u_i$ therefore gives us

$$\theta \leq \frac{u_i - x_i}{d_i}.$$

If $d_i = 0$, then x_i does not change and there is no restriction on θ .

So, our largest feasible step is

$$\theta^* = \min \left\{ u_j, \min_{i \in B: d_i < 0} \frac{x_i}{-d_i}, \min_{i \in B: d_i > 0} \frac{u_i - x_i}{d_i} \right\}.$$

The new basic columns depend on which term attains this minimum.

If $\theta^* = u_j$, then x_j reaches its lower bound before any basic variable reaches a bound. Thus, x_j moves from its upper bound u_j to its lower bound 0, and the basis does not change.

If the minimum is attained by some basic variable x_i , then x_i reaches either 0 or u_i and leaves the basis. The variable x_j becomes basic, so the column A_j replaces A_i in the basis.

(d)

Since every basic feasible solution is nondegenerate, every basic variable is strictly between its lower and upper bounds:

$$0 < x_i < u_i.$$

Therefore, all of the possible step sizes from parts (b) and (c) are strictly positive:

$$x_i > 0, \quad u_i - x_i > 0, \quad u_j > 0,$$

so the minimum of the possible step sizes also satisfies

$$\boxed{\theta^* > 0.}$$

We now show that the objective **strictly decreases** at every iteration.

In part (b), we choose $x_j = 0$ with $\bar{c}_j < 0$ and increase x_j by θ^* . Since

$$\begin{aligned} d_j &= 1 \quad \text{and} \quad d_B = -B^{-1}A_j, \\ c^T d &= c_j - c_B^T B^{-1}A_j = \bar{c}_j. \end{aligned}$$

Our total change in the objective is

$$\Delta z = \theta^* \bar{c}_j < 0,$$

since $\theta^* > 0$ and $\bar{c}_j < 0$.

Similarly, in part (c), we choose $x_j = u_j$ with $\bar{c}_j > 0$ and decrease x_j by θ^* . The change in the objective is now

$$\Delta z = -\theta^* \bar{c}_j < 0.$$

Thus, in either case, the objective strictly decreases at every iteration.

To show **termination**, we first establish that there are finitely many BFSs, then we show that each BFS can be visited at most once.

There are only finitely many possible basic feasible solutions. We can choose the m basic variables in

$$\binom{n}{m}$$

ways. Each of the remaining $n - m$ nonbasic variables can be at either its lower bound or its upper bound, so there are at most

$$\binom{n}{m} 2^{n-m}$$

possible basic configurations.

Since the objective strictly decreases at every iteration, we cannot return to a basic feasible solution that we have already visited. Therefore, since there are only finitely many basic feasible solutions, the algorithm must terminate after finitely many iterations.

Finally, we show that the solution **where the algorithm terminates is optimal**. Based on the definition of (b) and (c), when the algorithm stops, there is no variable in L with negative reduced cost and no variable in U with positive reduced cost. Therefore,

$$\bar{c}_j \geq 0 \quad \text{for } j \in L, \quad \bar{c}_j \leq 0 \quad \text{for } j \in U.$$

Now let y be any other feasible solution and let

$$d = y - x.$$

be the change in the variables needed to move from our current solution x to y .

We consider the two cases. If $j \in L$, then $x_j = 0$. Since y is feasible, we must have $y_j \geq 0$, so

$$d_j = y_j - x_j \geq 0.$$

Since the algorithm has stopped, we also have $\bar{c}_j \geq 0$. Therefore,

$$\bar{c}_j d_j \geq 0.$$

If $j \in U$, then $x_j = u_j$. Since y is feasible, we must have $y_j \leq u_j$, so

$$d_j = y_j - x_j \leq 0.$$

Since the algorithm has stopped, we also have $\bar{c}_j \leq 0$. Therefore,

$$\bar{c}_j d_j \geq 0,$$

Thus, for both $j \in L$ and $j \in U$, we have

$$\bar{c}_j d_j \geq 0.$$

Therefore, the total change in the objective between y and x is:

$$c^T y - c^T x = \sum_{j \in L} \bar{c}_j d_j + \sum_{j \in U} \bar{c}_j d_j \geq 0.$$

Hence,

$$\boxed{c^T y \geq c^T x.}$$

Since this holds for any feasible y , no feasible solution has a smaller objective value than x . Therefore, x is optimal. $\square$

Question 4: Production optimization

Weekly production and inventory planning over $T = 52$ weeks for $n = 20$ products made from $m = 500$ materials.

```
In [1]: using CSV, DataFrames, JuMP, Gurobi, Plots, Printf
        gr(); default(left_margin = 6Plots.mm, bottom_margin = 6Plots.mm)
```

Data

```
In [2]: datadir = joinpath(@__DIR__, "HW1_data")
        readmat(f) = Matrix{Float64}(CSV.read(joinpath(datadir, f), DataFrame)[: , 2:end])

        A = readmat("requirements.csv")
        b = readmat("availability.csv")
        d = readmat("demand.csv")
        p = CSV.read(joinpath(datadir, "profit.csv"), DataFrame).profit
        h = CSV.read(joinpath(datadir, "holding.csv"), DataFrame).holding

        m, n = size(A)
        T = size(d, 2)
        @assert size(b) == (m, T) && length(p) == n && length(h) == T
        (m = m, n = n, T = T)
```

```
Out[2]: (m = 500, n = 20, T = 52)
```

(a) Formulation

We have Products $i = 1, \dots, n$ ($n = 20$); materials $j = 1, \dots, m$ ($m = 500$); weeks $t = 1, \dots, T$ ($T = 52$).

We have the following data:

a_{ji} : how much of material j one unit of product i needs (from requirements.csv)

b_{jt} : how much of material j is available in week t (from availability.csv)

d_{it} : the demand for product i in week t (from demand.csv)

p_i is the unit profit of product i (from profit.csv)

h_t is the unit holding cost for inventory at the end of week t (from holding.csv).

We can either sell, make, or hold inventory of each product in each week. The goal is to maximize profit over the planning horizon (minus inventory holding costs).

Decision variables & Model

- $x_{it} \geq 0$: units of product i made in week t
- $s_{it} \geq 0$: units of product i sold in week t
- $I_{it} \geq 0$: units of product i in inventory at the end of week t , with $I_{i0} = 0$ (no starting stock)

$$\begin{aligned} \max_{x,s,I} \quad & \sum_{i=1}^n \sum_{t=1}^T p_i s_{it} - \sum_{i=1}^n \sum_{t=1}^T h_t I_{it} \\ \text{s.t.} \quad & \sum_{i=1}^n a_{ji} x_{it} \leq b_{jt} & \forall j, t & \quad (\text{weekly material availability, no carry-over}) \\ & s_{it} \leq d_{it} & \forall i, t & \quad (\text{cannot sell more than demand}) \\ & I_{i,t-1} + x_{it} - s_{it} = I_{it} & \forall i, t & \quad (\text{inventory balance, } I_{i0} = 0) \\ & x_{it}, s_{it}, I_{it} \geq 0 & \forall i, t & \end{aligned}$$

(b) Implementation

```
In [3]: model = Model(Gurobi.Optimizer)
        set_silent(model)
```

```

@variable(model, x[1:n, 1:T] >= 0) # production
@variable(model, s[1:n, 1:T] >= 0) # sales
@variable(model, I[1:n, 1:T] >= 0) # end-of-week inventory

@constraint(model, material[j = 1:m, t = 1:T], sum(A[j, i] * x[i, t] for i in 1:n) <= b[j, t])
@constraint(model, demand[i = 1:n, t = 1:T], s[i, t] <= d[i, t])
@constraint(model, balance[i = 1:n, t = 1:T], (t == 1 ? 0.0 : I[i, t-1]) + x[i, t] - s[i, t] == I[i, t])

@expression(model, sales_profit, sum(p[i] * s[i, t] for i in 1:n, t in 1:T))
@expression(model, holding_cost, sum(h[t] * I[i, t] for i in 1:n, t in 1:T))
@objective(model, Max, sales_profit - holding_cost)

optimize!(model)
@assert termination_status(model) == MOI.OPTIMAL
termination_status(model)

```

Set parameter Username

Set parameter LicenseID to value 2857457

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Out[3]: OPTIMAL::TerminationStatusCode = 1

In [4]: X, S, Inv = value.(x), value.(s), value.(I)
tol = 1e-6

```

@printf("Sales profit : \n", value(sales_profit))
@printf("Holding cost : \n", value(holding_cost))
@printf("Net profit : \n", objective_value(model))
println()
@printf("Units sold: %.1f of %.1f demanded (%.1f%%)\n", sum(S), sum(d), 100 * sum(S) / sum(d))

```

Sales profit : \$ 1370384.72

Holding cost : \$ 7371.37

Net profit : \$ 1363013.36

Units sold: 15807.4 of 246100.3 demanded (6.4%)

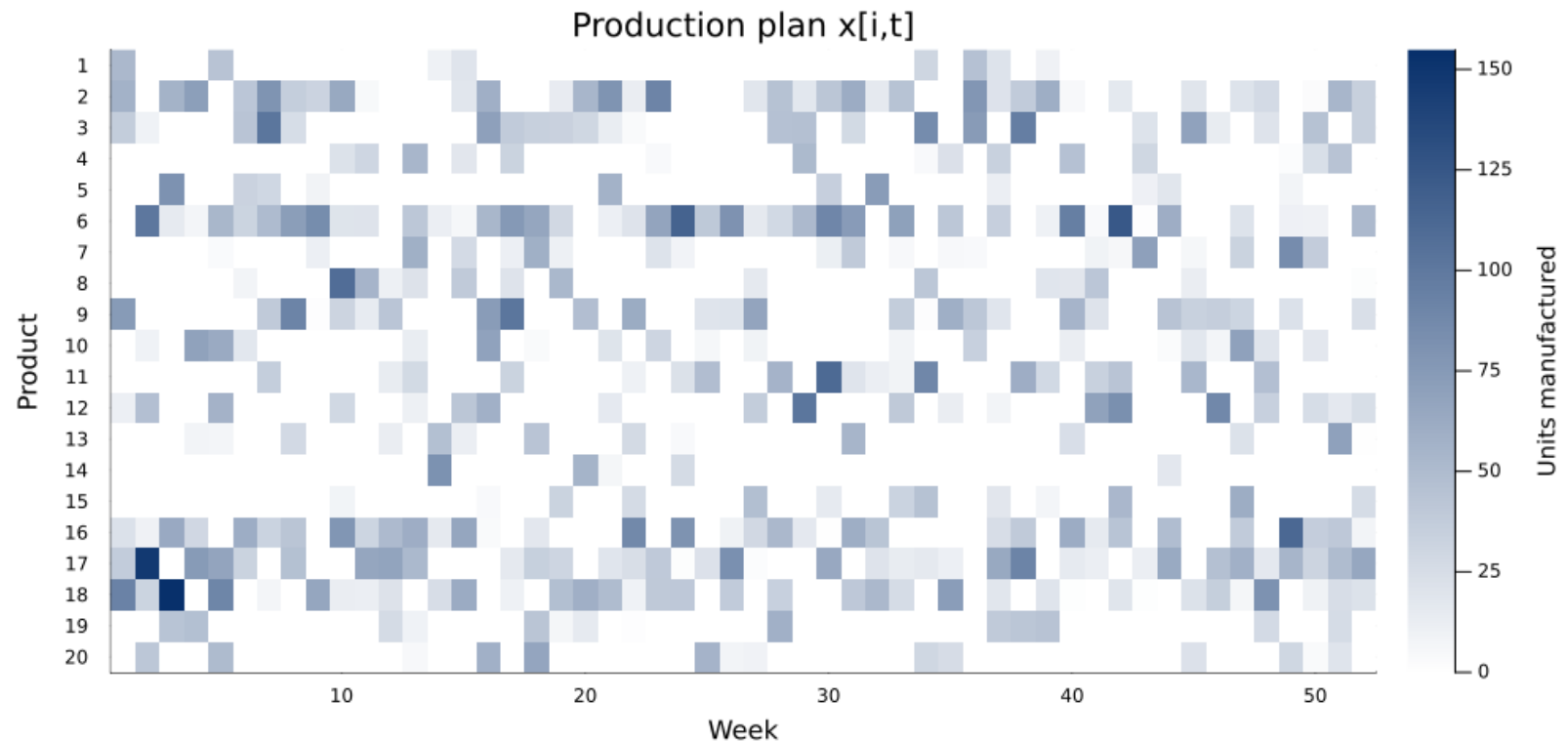
Result. The optimal solution yields a sales profit of **\$1,370,384.72** and a holding cost of **\$7,371.37**, corresponding to a net profit of **\$1,363,013.36**.

(c) Graphical analysis

(i) Units manufactured per product and week

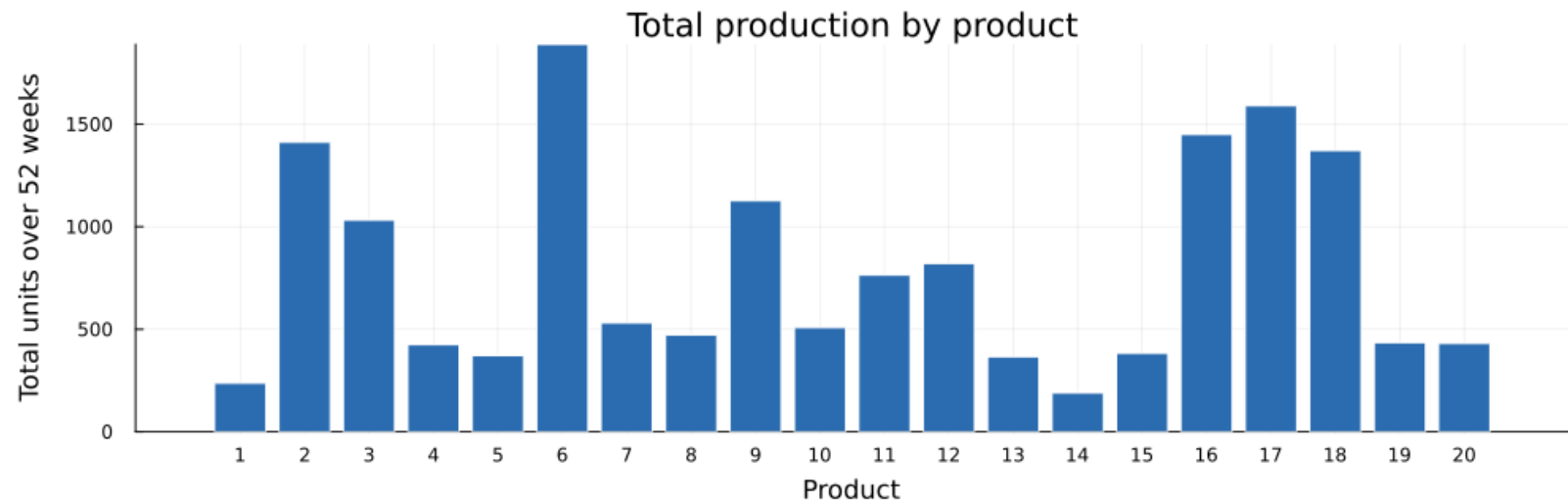
```
In [5]: seq = cgrad([:white, "#08306b"]) # single-hue sequential: white = 0
heatmap(1:T, 1:n, X;
        c = seq, clim = (0, maximum(X)),
        xlabel = "Week", ylabel = "Product", yticks = 1:n, yflip = true,
        colorbar_title = "Units manufactured",
        title = "Production plan x[i,t]", size = (1000, 480))
```

Out[5]:



```
In [6]: bar(1:n, vec(sum(X, dims = 2))); color = "#2b6cb0", linecolor = :white, legend = false,
        xlabel = "Product", ylabel = "Total units over 52 weeks", xticks = 1:n,
        title = "Total production by product", size = (1000, 320))
```

Out [6]:



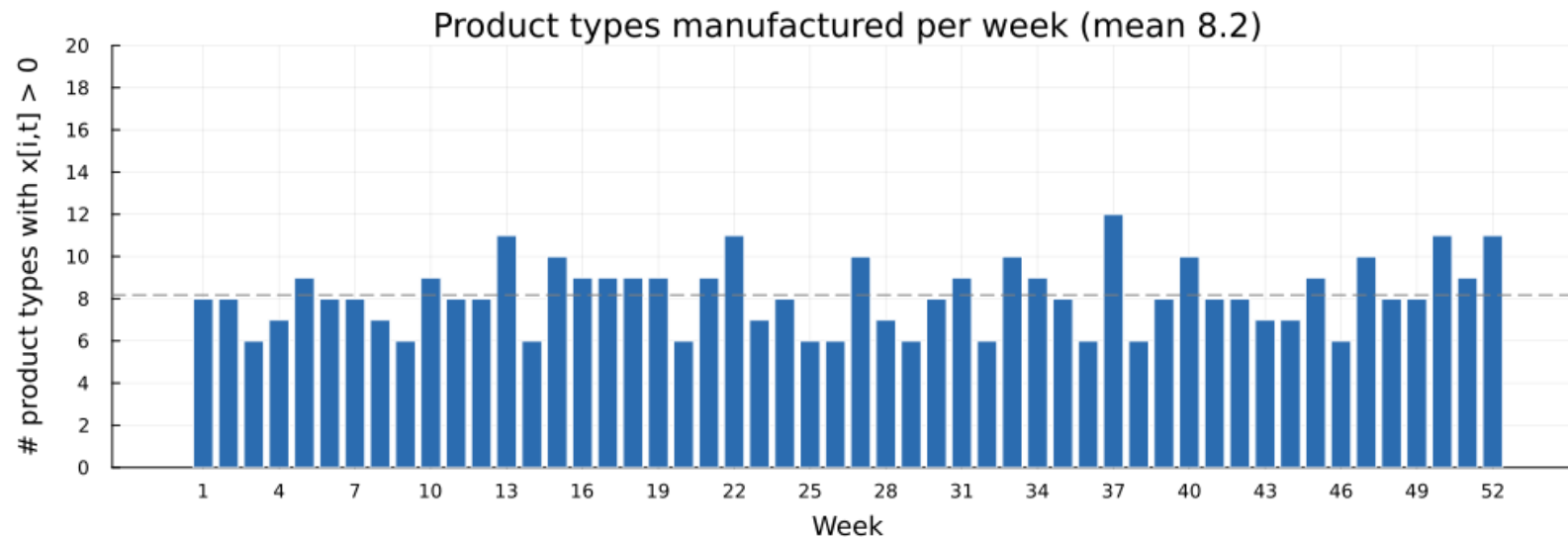
Interpretation. Production is relatively **sparse** and uneven across products and weeks. In most weeks, only a subset of the 20 products is manufactured, and the set of products produced changes substantially over time. Products 6, 16, 17, 18, 2, and 9 account for particularly large amounts of total production, while several other products are produced much less frequently. This variation occurs because material availability changes each week, so the optimal production plan changes accordingly.

(ii) Number of product types manufactured each week

```
In [7]: nprod      = [count(>(tol), X[:, t]) for t in 1:T]
nmat_bind = [count(A * X[:, t] .>= b[:, t] .- tol) for t in 1:T]
ndem_bind = [count(S[:, t] .>= d[:, t] .- tol) for t in 1:T]

bar(1:T, nprod; color = "#2b6cb0", linecolor = :white, legend = false,
    xlabel = "Week", ylabel = "# product types with x[i,t] > 0",
    ylims = (0, n), yticks = 0:2:n, xticks = 1:3:T,
    title = "Product types manufactured per week (mean $(round(sum(nprod)/T, digits=1)))",
    size = (1000, 340))
hline!([sum(nprod) / T]; color = :gray, linestyle = :dash)
```

Out[7]:



```
In [8]: println("Weeks where #products made == #binding materials: $(count(nprod .== nmat_bind)) / $T")
println("Every week, #made ≤ #binding materials + #demands met: $(all(nprod .<= nmat_bind .+ ndem_b"))
DataFrame{week = 1:T, products_made = nprod, materials_binding = nmat_bind, demands_met = ndem_bind}
```

Weeks where #products made == #binding materials: 47 / 52

Every week, #made ≤ #binding materials + #demands met: true

Out[8]: 52×4 DataFrame

27 rows omitted

Row	week	products_made	materials_binding	demands_met
	Int64	Int64	Int64	Int64
1	1	8	8	0
2	2	8	8	0
3	3	6	6	0
4	4	7	7	1
5	5	9	8	2
6	6	8	8	0
7	7	8	8	0

8	8	7	7	0
9	9	6	6	1
10	10	9	9	2
11	11	8	8	1
12	12	8	8	0
13	13	11	11	2
⋮	⋮	⋮	⋮	⋮
41	41	8	8	0
42	42	8	8	0
43	43	7	7	1
44	44	7	7	0
45	45	9	9	0
46	46	6	6	0
47	47	10	10	0
48	48	8	8	0
49	49	8	8	1
50	50	11	11	1
51	51	9	9	1
52	52	11	10	1

Interpretation. Between 6 and 12 of the 20 product types are manufactured each week, with an average of 8.2. Thus, the optimal solution generally concentrates production on only a subset of products rather than producing all 20 each week.

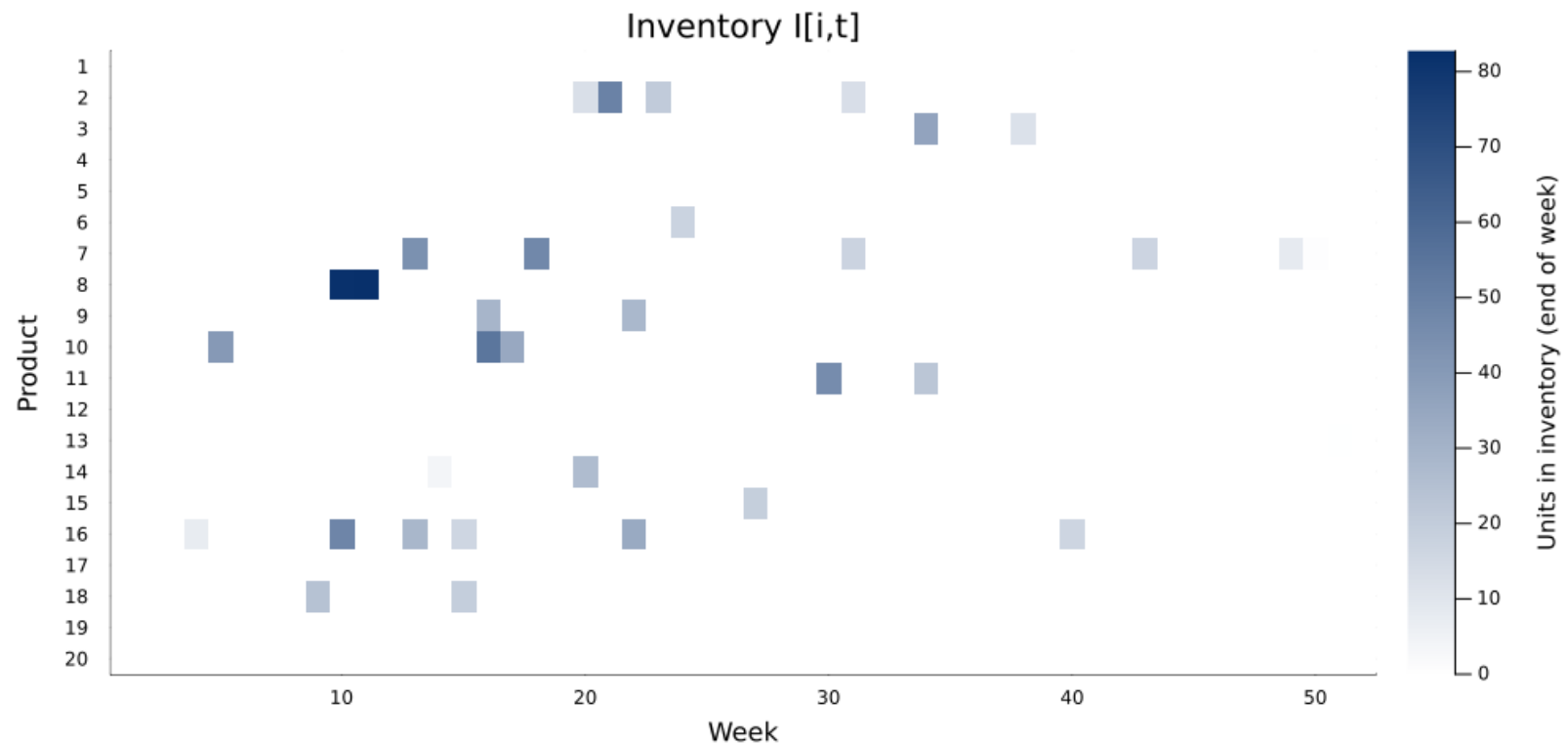
The number of products manufactured equals the number of binding material constraints in 47 of the 52 weeks. In

the remaining weeks, binding demand constraints also play a role. This suggests that weekly production is largely determined by a small set of scarce materials, and the relevant bottlenecks changes from week to week.

(iii) Units in inventory per product and week

```
In [9]: heatmap(1:T, 1:n, Inv;
             c = seq, clim = (0, maximum(Inv)),
             xlabel = "Week", ylabel = "Product", yticks = 1:n, yflip = true,
             colorbar_title = "Units in inventory (end of week)",
             title = "Inventory I[i,t]", size = (1000, 480))
```

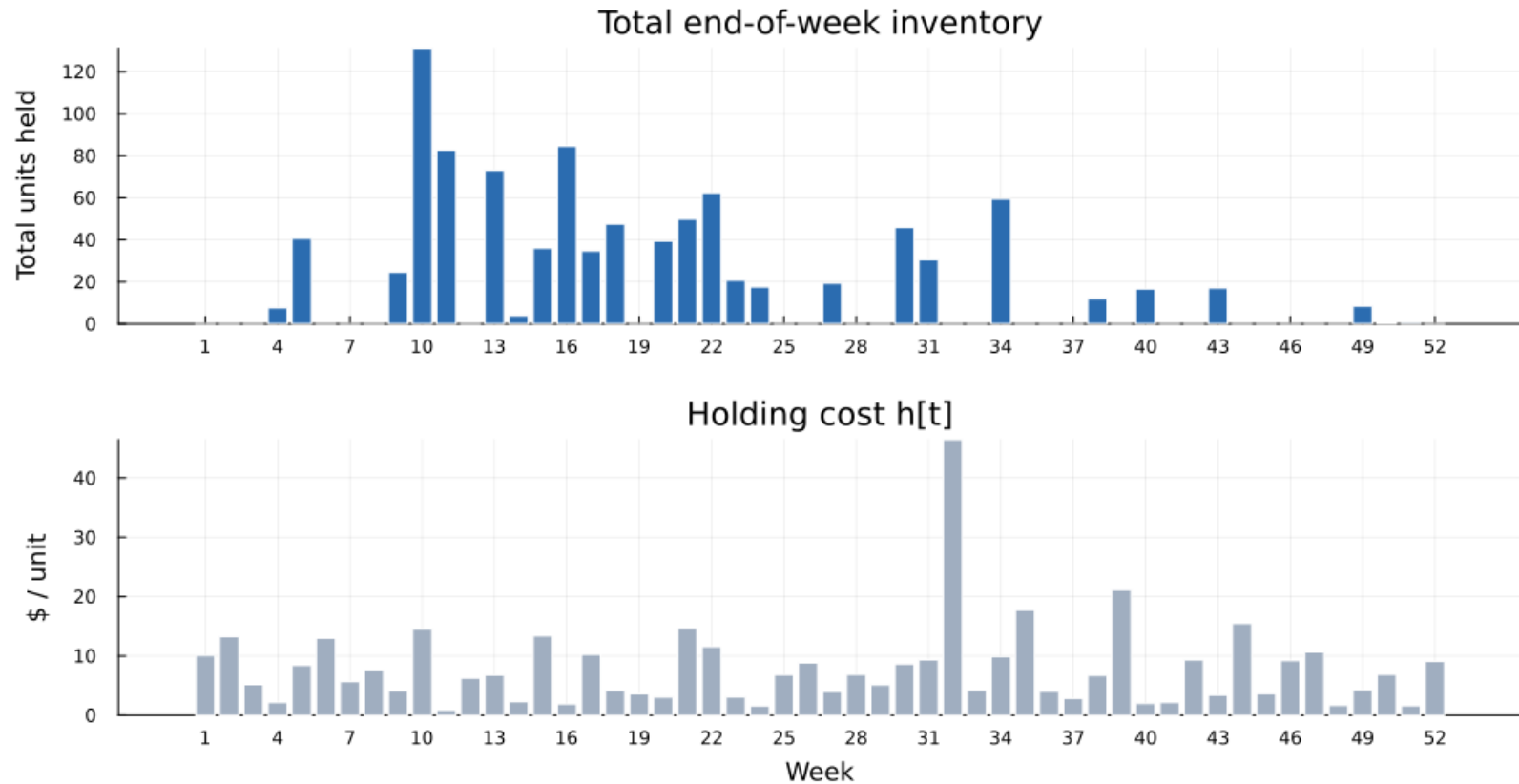
Out[9]:



```
In [10]: p1 = bar(1:T, vec(sum(Inv, dims = 1)); color = "#2b6cb0", linecolor = :white, legend = false,
                ylabel = "Total units held", title = "Total end-of-week inventory", xticks = 1:3:T)
p2 = bar(1:T, h; color = "#a0aec0", linecolor = :white, legend = false,
                xlabel = "Week", ylabel = "\$/ unit", title = "Holding cost h[t]", xticks = 1:3:T)
```

```
plot(p1, p2; layout = (2, 1), size = (1000, 520), link = :x)
```

Out[10]:



```
In [11]: held = [(i, t) for i in 1:n, t in 1:T if Inv[i, t] > tol]
prebuilt_undersaturated = [(i, t) for i in 1:n, t in 1:T if X[i, t] > tol && Inv[i, t] > tol && S[i, t] < tol]
multiweek = [(i, t) for i in 1:n, t in 2:T if Inv[i, t] > tol && Inv[i, t-1] > tol]
top5 = sortperm(h, rev = true)[1:5]
println("(product, week) cells with positive inventory: $(length(held)) of $(n * T)")
println("Stock built while that week's demand was NOT fully met: $(length(prebuilt_undersaturated))")
println("Units held for 2+ consecutive weeks (product, week): $multiweek")
println("Inventory at end of horizon: $(sum(Inv[:, T]))")
println("Total inventory in the 5 most expensive holding weeks $top5: $(round.(vec(sum(Inv, dims =
```

```
(product, week) cells with positive inventory: 34 of 1040
Stock built while that week's demand was NOT fully met: 0
Units held for 2+ consecutive weeks (product, week): [(8, 11), (10, 17), (2, 21), (7, 50)]
Inventory at end of horizon: 0.0
Total inventory in the 5 most expensive holding weeks [32, 39, 35, 44, 21]: [0.0, 0.0, 0.0, 0.0, 4
9.9]
```

Interpretation. Inventory is relatively sparse: **only 34 of the 1,040 product-week combinations have positive inventory**, and inventory is generally held for only a short period.

This suggests that the model primarily produces goods when they are needed rather than storing them for long periods, since holding inventory incurs an additional cost. Inventory is used occasionally to shift production across weeks. For example, producing extra units in a week when materials are available and carrying them into a later week when production is more constrained. We also see that no inventory remains at the end of week 52.

(d) Week 30

```
In [12]: t0 = 30

use = A * X[:, t0]

full_mat = findall(use .>= b[:, t0] .- tol)
met_dem = findall(S[:, t0] .>= d[:, t0] .- tol)

println("Materials fully utilized: ", length(full_mat), " of 500")
println("Materials: ", full_mat)

println("Products with demand fully met: ", length(met_dem), " of 20")
println("Products: ", met_dem)
```

```
Materials fully utilized: 8 of 500
Materials: [16, 42, 48, 141, 190, 201, 354, 484]
Products with demand fully met: 1 of 20
Products: [11]
```

Answer. In week 30, **8 of the 500 materials are fully utilized** (materials 16, 42, 48, 141, 190, 201, 354, and 484),

while demand is fully satisfied for only **1 of the 20 products** (product 11).

This means that 8 material availability constraints are binding, compared with only one demand constraint. Overall, production in week 30 is mainly limited by material availability rather than demand. Most products still have unmet demand, but producing more of them would require materials that are already fully utilized.